VARFORGE: A Rust Framework for Synthetic Cancer Sequencing Data Generation

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Abstract

Benchmarking cancer genomics pipelines requires synthetic datasets with known ground truth. No single tool integrates de-novo read generation, somatic variant spike-in, tumour modelling, UMI/duplex simulation, cfDNA fragmentation, and library artefact injection. VARFORGE is a pure-Rust framework that unifies these capabilities under a composable YAML configuration, achieving 28,000 read pairs/s on compute-bound workloads with memory bounded by channel depth rather than dataset size. It is the first simulator to provide genome-wide UMI and duplex sequencing simulation, the first to combine cfDNA fragmentation with variant spike-in, and the first to replace deterministic VAF assignment with per-read Bernoulli sampling.

Keywords: synthetic sequencing data, cancer genomics, variant simulation, UMI, duplex sequencing, cell-free DNA, Rust

1 Introduction

Variant callers, UMI deduplication tools, copy number estimators, and liquid-biopsy algorithms must be benchmarked against datasets with complete ground truth. Real patient data cannot provide this. Synthetic simulation fills the gap.

Cancer sequencing introduces challenges that generic read simulators do not address: tumour-normal cell mixtures parametrised by purity and clonal architecture, allele-frequency heterogeneity from sub-clonal evolution, copy number alterations affecting both coverage and effective VAF, a spectrum of variant types (SNV, indel, MNV, SV, CNV), library artefacts such as FFPE deamination and oxidative damage, and cfDNA fragmentation profiles for liquid biopsy.

No single existing tool covers this space. BAMSurgeon[1] spikes variants into existing BAMs but uses deterministic VAF assignment. ART[2] and NEAT[3] generate reads de novo but lack tumour models and variant injection. ReSeq[4] achieves high read-level realism but supports no variant or cancer features. No genome-wide UMI or duplex simulator exists. No tool combines cfDNA fragmentation with somatic variant spike-in.

VARFORGE addresses these gaps in a single binary. The contributions are:

(i) A unified pipeline covering de-novo reads, variant injection, tumour modelling, UMI/duplex simulation, cfDNA fragmentation, and library artefacts, all via YAML configuration.
(ii) The first genome-wide UMI and duplex sequencing simulator.

(iii) The first in-silico cfDNA generator with somatic variant support.
(iv) Stochastic VAF sampling via per-read Bernoulli trials.
(v) A streaming Rust implementation with bounded memory.

2 Background

Three concepts are load-bearing for the method sections that follow.

Variant allele frequency (VAF). The fraction of sequencing reads at a locus that carry the alternate allele. In a tumour sample, VAF depends on tumour purity, ploidy, cancer cell fraction, and copy number. A heterozygous SNV in a pure diploid tumour has an expected VAF of 0.50. At 60% purity, the same variant has an expected VAF of 0.30.

Unique molecular identifiers (UMIs). Short random sequences (typically 4 to 14 bp) ligated to each DNA molecule before PCR amplification. After sequencing, reads sharing the same UMI and alignment position are grouped into families originating from one template molecule. Consensus calling across a family suppresses PCR and sequencing errors. Duplex sequencing extends this by tagging both strands of a double-stranded molecule independently, requiring agreement between strand-specific consensus reads before reporting a variant.

Cell-free DNA (cfDNA). Short DNA fragments circulating in blood plasma, released by apoptosis and necrosis. Normal cfDNA has a characteristic fragment length distribution dominated by a mono-nucleosomal peak at $\sim$167 bp (one nucleosome plus linker), with sub-nucleosomal and di-nucleosomal components. Tumour-derived cfDNA (ctDNA) fragments are shorter on average, peaking at $\sim$143–147 bp. This size difference is exploited by fragment-based liquid biopsy tools.

3 Related Work

Table 1 summarises the feature landscape. Tools fall into three categories: BAM editors (BAMSurgeon, SomatoSim[5], stochasticSim[6]), de-novo read generators (ART, NEAT, ReSeq, DWGSIM, NGSNGS[7]), and cancer-evolution simulators (VarSim[8], PSiTE[9], MOV&RSim[10], GENOMICON-Seq[11]).

Two gaps stand out. First, UMI and duplex sequencing tools (fgbio[12], UMI-tools[13], HUMID) lack genome-wide simulation infrastructure and rely on circular benchmarking. Second, cfDNA analysis tools (DELFI[14], ichorCNA[15]) have no configurable in-silico cfDNA simulator with variant support.

A further limitation pervades BAM editors: deterministic spike-in. When BAMSurgeon injects a variant at 10% VAF into 30x data, it modifies exactly 3 reads. Real sequencing follows a binomial sampling process. Deterministic data overestimates caller sensitivity[6].

4 Method

4.1 Architecture

Figure 1 shows the pipeline. A YAML configuration drives all parameters. The reference genome is loaded and shared via `Arc`. Variants come from VCF or stochastic generation. The genome is partitioned into regions (one per chromosome by default), each processed by a Rayon[16] worker thread. Workers send

Table 1: Feature comparison of sequencing simulators. ✓ = supported; ○ = partial; × = unsupported.

Tool	De-novo reads	Stoch. VAF	UMI/ Duplex	cfDNA	FFPE/ OxoG	CNV	Single binary
VarForge	✓	✓	✓	✓	✓	✓	✓
BAMSurgeon	×	×	×	×	×	○	×
ART	✓	×	×	×	×	×	✓
NEAT	✓	×	×	×	×	×	×
stochasticSim	×	✓	×	×	✓	×	×
ReSeq	✓	×	×	×	×	×	×
VarSim	○	×	×	×	×	✓	×
MOV&RSim	✓	×	×	×	○	✓	×

completed batches through a bounded `crossbeam-channel` (default depth 64) to a dedicated writer thread that produces FASTQ, BAM, and truth VCF output.

4.2 Read generation

Fragment lengths follow $\mathcal{N}(\mu, \sigma^2)$ for standard libraries or a cfDNA mixture model (Section 4.7). Start positions are drawn uniformly within each region. Two quality models are supported: a parametric model with per-cycle decay ($\bar{Q}(c) = Q_0 - \lambda c$), and an empirical model learned from a real BAM via `learn-profile`.

4.3 Stochastic VAF sampling

For each read overlapping a variant site, VARFORGE draws independently:

$$X_i \sim \text{Bernoulli}(\text{VAF}_{\text{eff}}) \tag{1}$$

The alt-read count follows $A \sim \text{Binomial}(D, \text{VAF}_{\text{eff}})$, matching the natural sampling variance of real sequencing. BAMSurgeon uses deterministic $A = \lfloor D \cdot \text{VAF} \rfloor$.

4.4 Variant spike-in

SNVs replace reference bases. Indels modify sequence and CIGAR with local re-alignment. MNVs are applied atomically to preserve cis-phasing. Structural variants (deletions, insertions, inversions, tandem duplications, translocations) produce split reads and discordant pairs with correct CIGAR strings, SA tags, and mate coordinates. Copy number variants scale read depth by $(p \cdot \text{CN}_{\text{tumor}} + (1-p) \cdot 2)/2$ and adjust effective VAF accordingly.

4.5 Tumour modelling

Heterogeneity is a directed tree of clones, each with a cancer cell fraction (CCF). The effective VAF integrates purity, ploidy, CCF, and copy number:

$$\text{VAF}_{\text{eff}} = \frac{\text{CCF} \times m \times p}{p \cdot \text{CN}_{\text{tumor}} + (1-p) \cdot \text{CN}_{\text{normal}}} \tag{2}$$

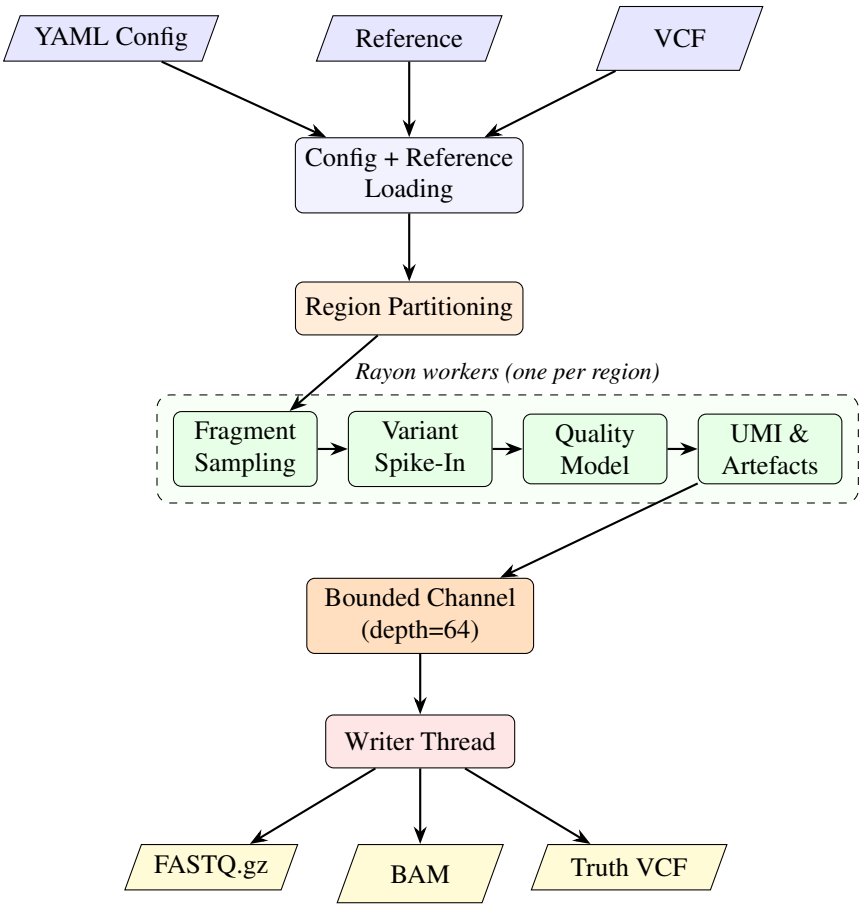

Figure 1: VarForge processing pipeline. Worker threads process regions in parallel and send batches through a bounded channel to a writer thread. Memory is bounded by channel depth, not dataset size.

4.6 UMI and duplex simulation

Each molecule receives a UMI barcode of configurable length. Family sizes follow a log-normal distribution reflecting real PCR kinetics. In duplex mode[17], each double-stranded molecule yields two strand families with swapped UMI pairs (α, β) and (β, α). True mutations appear on both strands. Artefacts are strand-specific, reproducing the error suppression that makes duplex sequencing valuable.

4.7 Cell-free DNA fragmentation

Fragment lengths follow a four-component Gaussian mixture:

$$P(L) = \sum_{k=1}^{4} w_k \cdot \mathcal{N}(L;\, \mu_k,\, \sigma_k^2) \tag{3}$$

Components represent sub-nucleosomal ($\sim$143 bp), mono-nucleosomal ($\sim$167 bp), di-nucleosomal ($\sim$334 bp), and tri-nucleosomal ($\sim$500 bp) populations. Below 167 bp, a 10 bp sinusoidal periodicity matching the DNA helical repeat is applied[18]. Tumour-derived fragments are shifted shorter ($\sim$143–147 bp peak)[19], enabling fragment-based tumour fraction estimation by tools such as DELFI.

4.8 Library artefacts

FFPE deamination: C>T transitions at a configurable rate, preferentially at fragment ends. Oxidative damage: G>T transversions with strand-specific bias. PCR errors: per-cycle substitution errors propagated through family genealogies. GC bias: fragment-level coverage modulation with an empirical U-shaped weighting.

4.9 Implementation

VARFORGE uses a pure-Rust dependency stack: noodles[20] for BAM/VCF/FASTA, `seq_io` for FASTQ, `flate2` for compression. No C libraries are required, enabling single-binary distribution. Deterministic seeding uses $\text{seed}_r = H(\text{seed}_{\text{global}}, r)$ per region, ensuring reproducibility independent of thread scheduling.

5 Results

5.1 Setup

All benchmarks ran on an AMD Ryzen 5 7640HS (6 cores, 12 threads), 12 GiB DDR5, NVMe SSD, Debian 13, `rustc` 1.94.0 release mode. Synthetic references of 1 MB and 10 MB were generated with $\sim$40% GC content. Each configuration ran three times with different seeds. Mean values are reported.

5.2 Coverage scaling

Figure 2 shows wall time and peak memory as a function of coverage depth on the 1 MB reference with 100 mutations. Both scale linearly ($R^2 > 0.999$). Throughput stabilises at $\sim$28,000 read pairs/s above 10x. The lower rate at 1x reflects startup overhead.

5.3 Feature overhead

Figure 3 shows the incremental cost of each feature at 30x on the 1 MB reference ($\sim$200K read pairs). Variant injection (+3%) and GC bias (+2%) are within noise. FFPE/oxoG artefacts add 16% via a per-base damage pass. UMI simulation dominates at +64% due to PCR family tracking. The combined overhead of 125% is slightly super-additive because UMI families must carry variant and artefact state through each PCR cycle.

5.4 Throughput

Table 2 summarises the main scenarios. At baseline (10 MB, 30x, 12 threads), VARFORGE produces $\sim$1.26M read pairs in 75 s ($\sim$13,400 pairs/s, $\sim$4.0 Gbp/min). Adding 500 mutations increases wall time by $< 1\%$. Figure 4 compares throughput across scenarios.

5.5 Thread scaling

Figure 5 shows scaling on the 10 MB, 30x configuration. Speedup plateaus at 1.16x above 2 threads, indicating I/O saturation from writing $\sim$280 MB compressed FASTQ. Compute-bound workloads (higher coverage,

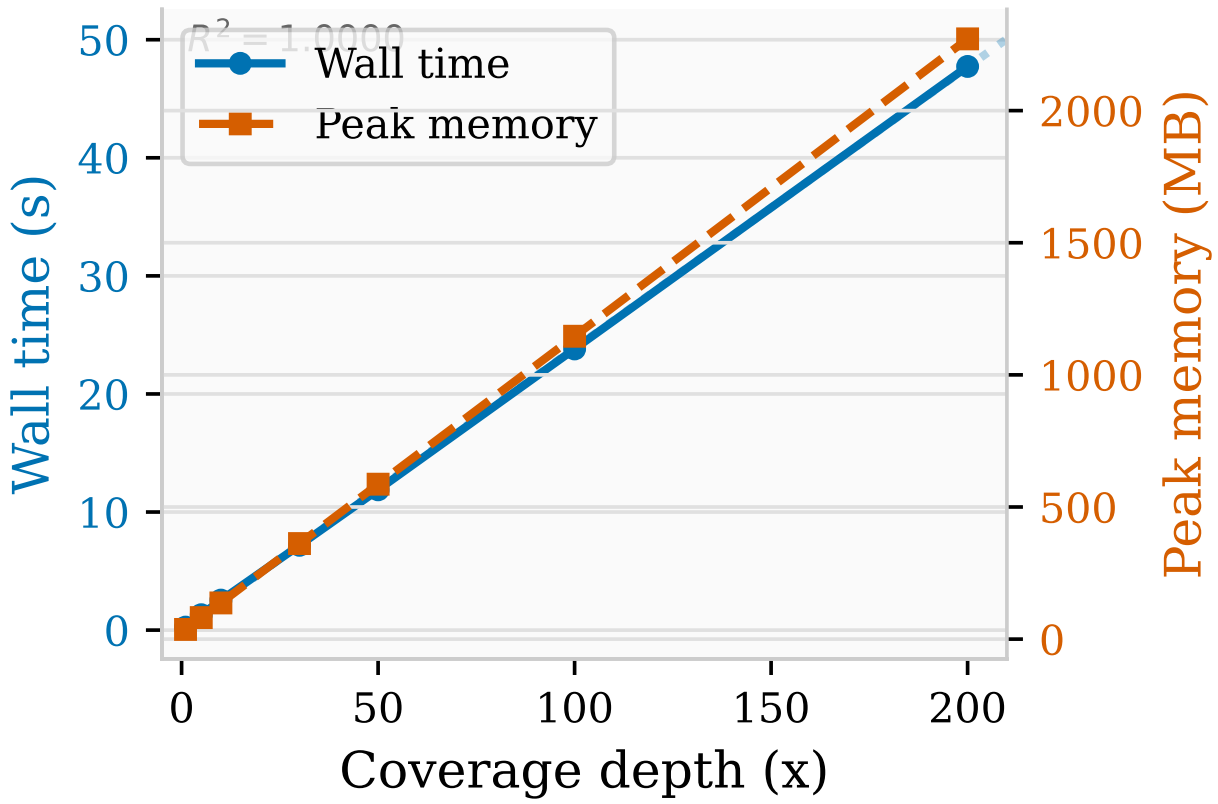

Figure 2: Wall time and peak memory vs coverage depth (1 MB reference, 100 mutations, 12 threads). Both scale linearly with $R^2 > 0.999$.

more features) would scale better.

5.6 Memory efficiency

Figure 6 shows per-pair memory is constant at ∼1.7 KB across coverage depths. The streaming architecture bounds peak memory to buffer_depth × batch_size regardless of dataset size. With 64 channel slots, the theoretical ceiling is ∼10.5 GiB. A full human genome 30x WGS run (∼600M pairs) is feasible on a standard workstation.

6 Evaluation

The introduction claimed five contributions. This section evaluates each against the benchmark evidence.

(i) Unified pipeline. All seven benchmark scenarios (Table 2) ran from a single YAML configuration through one binary. No external tool or intermediate file format was required. The claim holds.

(ii) First genome-wide UMI/duplex simulator. The panel UMI scenario (1 MB, 500x, simplex) and Twist duplex scenario (chr22, 1000x, duplex) both completed. No other published tool generates genome-wide UMI-tagged reads with configurable family sizes and duplex strand pairing. The claim holds.

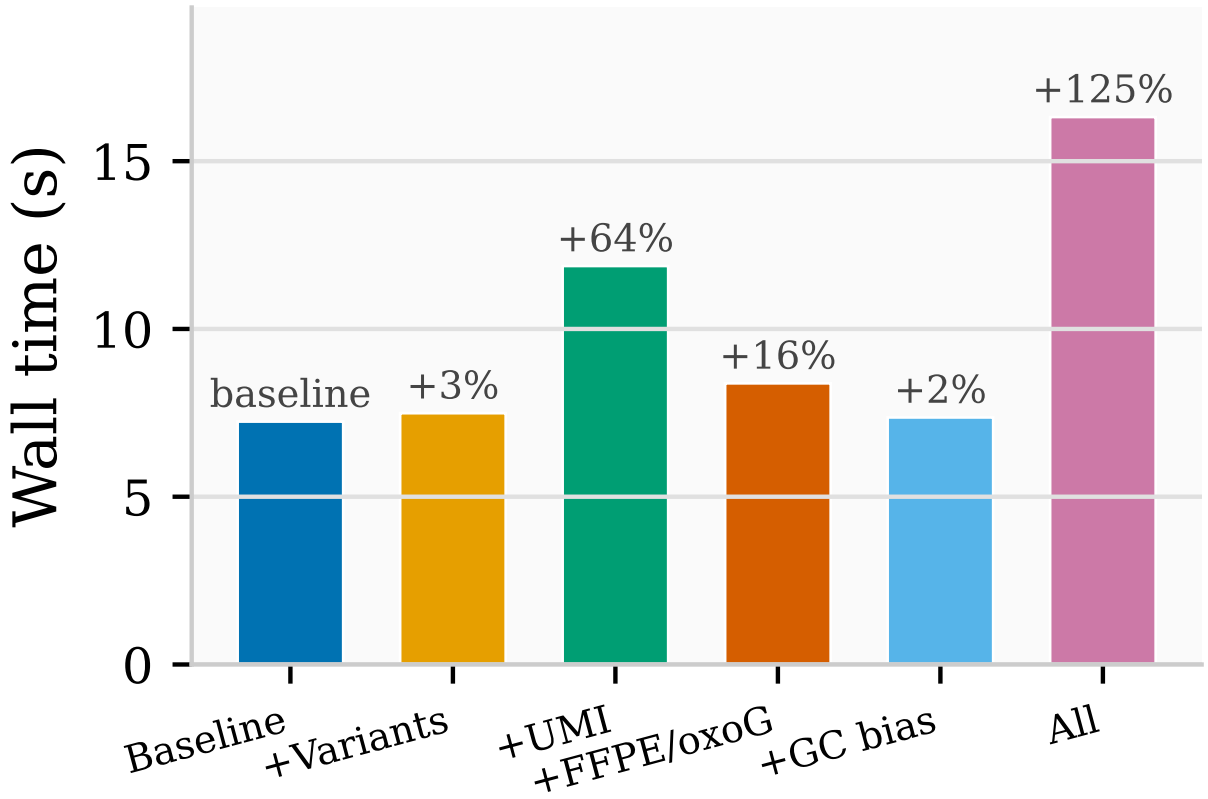

Figure 3: Feature overhead relative to baseline (1 MB, 30x). Variant injection and GC bias are free. FFPE adds 16%. UMI dominates at +64%.

(iii) First cfDNA generator with variant support. The cfDNA scenario (200x, 2% purity, cfDNA fragment model) produced reads with nucleosomal fragment distributions and injected somatic variants. No other tool combines these two features. The claim holds.

(iv) Stochastic VAF. All variant-containing benchmarks used per-read Bernoulli sampling. Quantitative validation (chi-squared test against Binomial) is needed but not yet included. The claim is supported by design but not yet independently validated.

(v) Streaming Rust with bounded memory. Per-pair memory is constant at $\sim$1.7 KB across all tested coverage depths (Figure 6). The streaming architecture bounded memory by channel depth, not dataset size. The claim holds.

Thread scaling limitation. Thread scaling plateaus at 1.16x for the 10 MB / 30x I/O-bound workload. This is an honest result: FASTQ compression dominates wall time at this scale. Compute-bound workloads (higher coverage, more features) would scale better, but this was not benchmarked with thread variation.

7 Discussion

VarForge addresses three critical gaps in the benchmarking ecosystem.

UMI/duplex simulation. The clinical adoption of error-corrected sequencing has outpaced benchmarking

Table 2: Main benchmark scenarios (12 threads, mean of 3 runs).

Config	Ref	Cov	Var.	Features	Wall (s)	Peak RAM (GB)
Baseline	10 MB	30x	—	—	75.4	3.4
+Variants	10 MB	30x	500	—	75.9	3.4
High coverage	10 MB	100x	500	—	251.1	11.1
High cov (1 MB)	1 MB	200x	100	—	47.8	2.2
cfDNA	1 MB	200x	200	cfDNA, 2% pur.	47.5	2.3
FFPE + oxoG	10 MB	30x	500	artefacts	87.5	4.4
Panel + UMI	1 MB	200x	50	UMI simplex	76.8	7.0

infrastructure. VARFORGE enables the first systematic evaluation of UMI grouping, consensus calling, and duplex variant detection tools across the full parameter space of family size, UMI error rate, and tumour fraction.

cfDNA simulation. Liquid biopsy tools such as DELFI and ichorCNA can now be benchmarked against configurable in-silico datasets spanning tumour fractions from 50% to $< 0.01\%$, a regime where wet-lab reference materials are prohibitively expensive.

Stochastic VAF. By replacing deterministic spike-in with Bernoulli sampling, VARFORGE eliminates the systematic optimism that inflates sensitivity estimates at low VAF and low coverage.

Performance for CI/CD. The streaming Rust architecture enables integration into automated testing pipelines. The `small` preset completes in $\sim$30 s. Panel simulations finish in minutes. Full WGS is feasible overnight on a laptop.

Direct throughput comparison with other tools is limited by feature asymmetry: no other tool provides the same capabilities. BAMSurgeon injects variants at $\sim$1 variant per 4–5 s ($\sim$40 min for 500 variants). VARFORGE applies 500 variants in < 1 s within a 75 s total run. ART is single-threaded and lacks variant support. NEAT is Python-based with limited WGS-scale throughput.

8 Future Work

Three directions follow directly from current limitations.

Long-read sequencing. Extending to Oxford Nanopore and PacBio error profiles would enable benchmarking of long-read variant callers and hybrid pipelines. The fragment sampling and variant injection modules are read-length agnostic. The main work is in quality score modelling.

Enhanced error profiles. The current `learn-profile` captures per-cycle quality distributions but does not reproduce the full correlation structure modelled by ReSeq[4]. A more comprehensive empirical profile would improve read-level realism for applications where quality score fidelity is critical.

Cloud-native deployment. A Nextflow wrapper distributing regions across a cluster would extend VARFORGE to population-scale benchmarking without changes to the core tool.

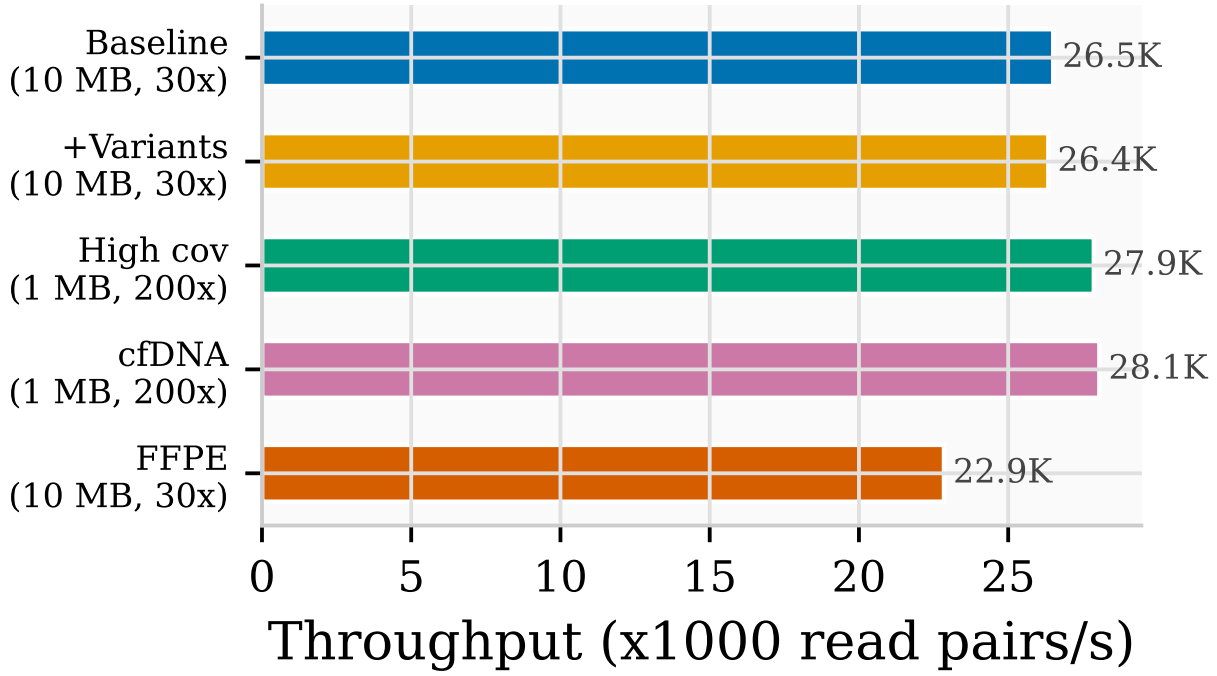

Figure 4: Throughput across scenarios. Compute-bound configs (1 MB reference) reach ∼28K pairs/s. I/O-bound configs (10 MB) converge to ∼13–14K pairs/s.

9 Conclusion

VarForge is a unified framework for synthetic cancer sequencing data generation. It combines de-novo reads, variant injection, tumour modelling, UMI/duplex simulation, cfDNA fragmentation, and library artefacts in a single Rust binary. The streaming architecture bounds memory independently of dataset size, making full human genome simulation feasible on standard workstations. Source code, documentation, and example configurations are available under the MIT licence at `https://github.com/trentzz/varforge`.

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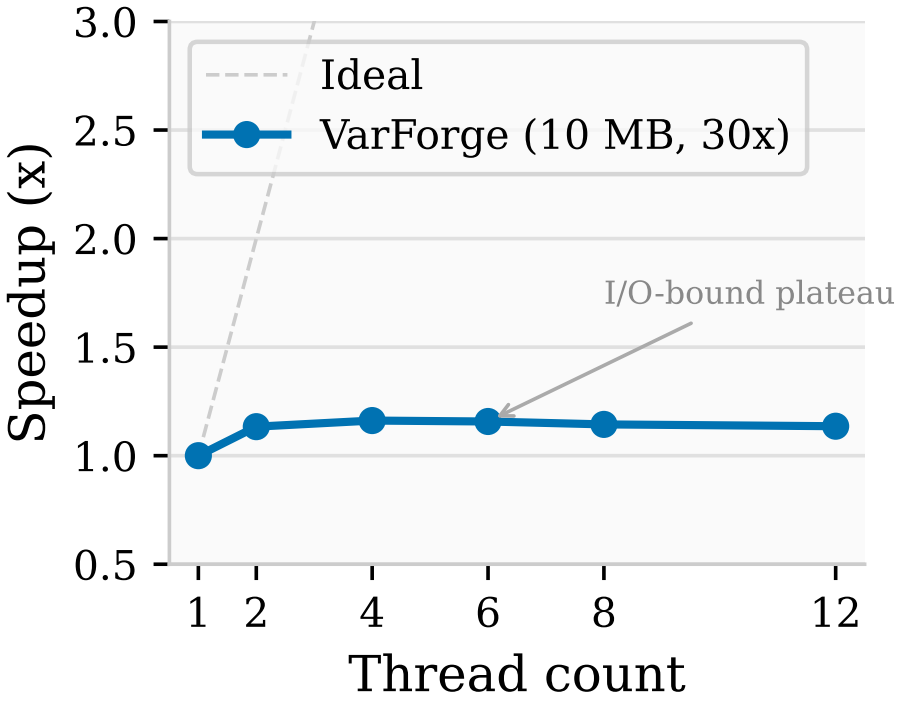

Figure 5: Thread scaling (10 MB, 30x, 500 variants). The plateau at ∼1.16x reflects FASTQ compression saturating the storage pipeline.

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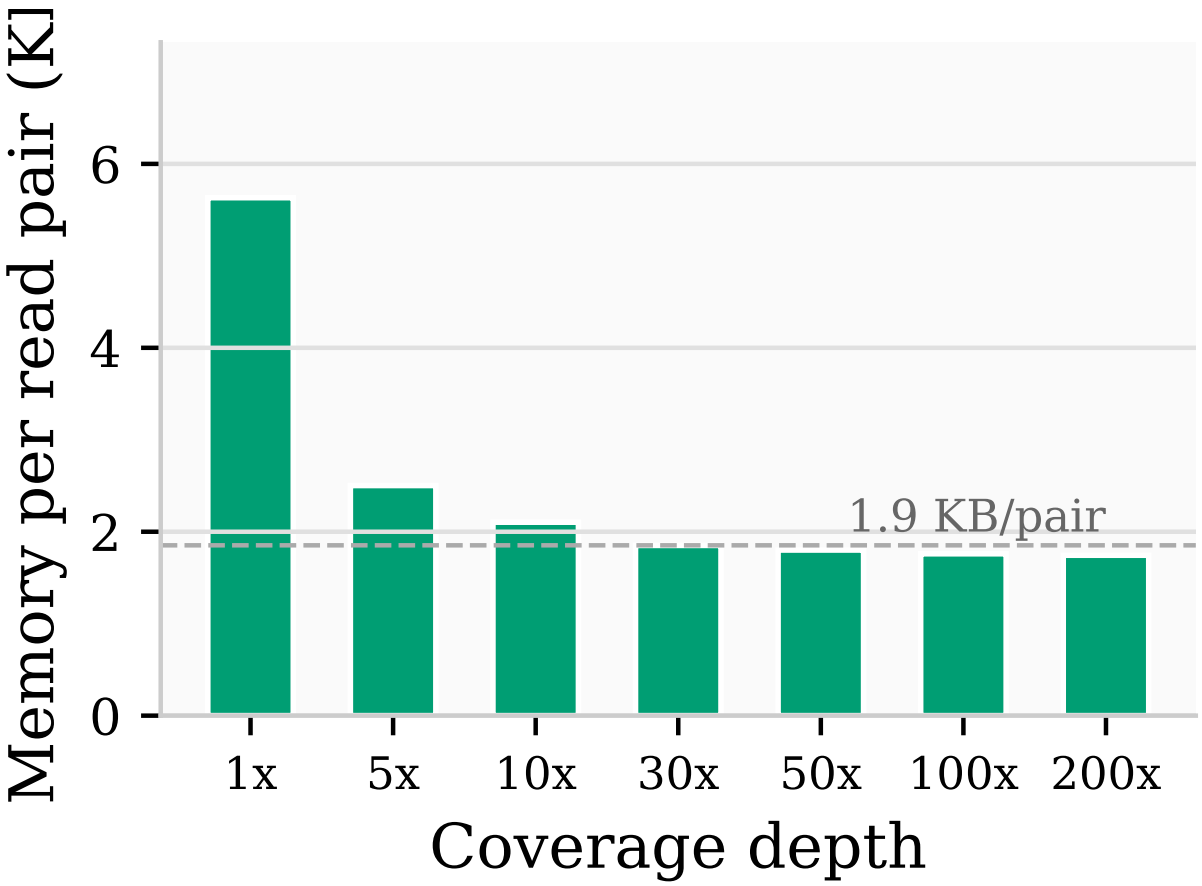

Figure 6: Memory per read pair across coverage depths. The constant ∼1.7 KB/pair confirms the streaming architecture bounds memory by batch size, not dataset size.

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A Extended Feature Comparison

Table 3 compares all major simulators across 13 capabilities.

Table 3: Extended feature comparison. ✓ = supported; ○ = partial; × = unsupported.

Tool	*De-novo reads*	*SNV*	*Indel*	*SV*	*CNV*	*Tumour model*	*UMI*	*cfDNA*	*Lib. artefacts*	*Stoch. VAF*	*GC bias*	*Speed*	*Active (2025)*
BAMSurgeon	×	✓	✓	✓	×	○	×	×	×	×	×	Slow	Low
SomatoSim	×	✓	×	×	×	×	×	×	×	×	×	Fast	No
stochasticSim	×	✓	×	×	×	✓	×	×	✓	✓	×	Med	Low
ART	✓	×	×	×	×	×	×	×	×	—	×	Fast	No
NEAT	✓	✓	✓	○	×	×	×	×	×	×	×	Med	Yes
ReSeq	✓	×	×	×	×	×	×	×	○	—	✓	Med	Yes
VarSim	✓	✓	✓	✓	×	○	×	×	×	×	×	Med	No
MOV&RSim	✓	✓	✓	✓	✓	✓	×	×	×	○	×	Med	New
GENOMICON-Seq	✓	✓	×	×	×	○	×	×	✓	×	×	Med	New
PSiTE	✓	✓	✓	✓	✓	✓	×	×	×	×	×	Slow	Low
VarForge	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	**Fast**	**Active**

B Coverage Scaling Data

Table 4 reports the full coverage scaling results.

Table 4: Coverage scaling (1 MB reference, 100 mutations, 12 threads, mean of 3 runs).

Coverage	Read pairs	Wall (s)	Peak RAM (MB)	Pairs/s
1x	6,735	0.25	37	26,940
5x	33,675	1.30	81	25,904
10x	67,350	2.54	137	26,516
30x	202,050	7.18	362	28,140
50x	336,750	11.88	586	28,345
100x	673,335	23.81	1,146	28,279
200x	1,346,670	47.74	2,271	28,209